

Excursus: Financial Statement Analysis for the German Stock Market

- 1) Case Study: Wirecard**
 - 2) Empirical analysis – Fundamental Analysis for publicly listed German stocks, 1995-2021**
- 2 Valuation Metrics:**
- M/B Ratio**
 - F-Score (Piotroski, 2000)**

■ Background:

- Founded in Germany in 1999, Wirecard AG operated in the digital payment segment of the FinTech sector. The shares of Wirecard were listed in the DAX. Its market capitalisation was around € 24 billion in September 2018.¹
- On 25 June 2020, Wirecard AG filed for insolvency. The week before, EY, Wirecard's auditor, had announced that 1.9 billion euros, roughly a quarter of Wirecard's balance sheet, were missing from its accounts. Earlier in 2020, a special audit by KPMG had questioned the accuracy of Wirecard's financial reporting and raised serious doubts about the existence of large parts of Wirecard's profits reported between 2016 and 2018.²
- Did Financial Statement Analysis help in detecting potential fraud?

1. Zeranski and Sancak, 2020, The Wirecard Scandal: The High-speed Rise and Fall of a FinTech Company and Its Implications for Developed and Developing Economies. <https://www.law.ox.ac.uk/business-law-blog/blog/2020/10/wirecard-scandal-high-speed-rise-and-fall-fintech-company-and-its>.
2. Jakubeit, 2021, The Wirecard scandal and the role of BaFin. <https://sep.luiss.it/sites/sep.luiss.it/files/WP5.21%20The%20Wirecard%20scandal%20and%20the%20role%20of%20Bafin.pdf>.

■ Red Flag: Borrowing for no clear purpose

- The total current interest-bearing assets were increased by 46%, 25%, 35%, 38% for years 2015, 2016, 2017, and 2018.

In EUR million	2014	2015	2016	2017	2018
(1) Interest-bearing securities and fixed-term deposits	122,9	133,1	156,5	109,1	139,6
(2) Cash and cash equivalents	695,1	1.063,0	1.332,6	1.901,3	2.719,8
(3) Total current (liquid) interest-bearing assets [= (1) + (2)]	818,0	1.196,1	1.489,1	2.010,4	2.859,4
(4) Total interest-bearing liabilities (non-current and current)	98,4	370,7	594,5	1.066,4	1.466,1
(5) Total assets	1.995,2	2.935,5	3.482,1	4.532,8	5.854,9
SELECTED ACCOUNTING RATIOS:					
Liquid interest-bearing assets / Total assets [= (3) / (5)]	41,0%	40,7%	42,8%	44,4%	48,8%
Interest-bearing liabilities / Total assets [= (4) / (5)]	4,9%	12,6%	17,1%	23,5%	25,0%
Interest-bearing liabilities / Liquid interest-bearing assets [= (4) / (3)]	12,0%	31,0%	39,9%	53,0%	51,3%

Source: Financial Times, 2020, Spotting red flags: Wirecard edition, <https://www.ft.com/content/f70686c7-473c-4a5b-90b0-89094b534394>

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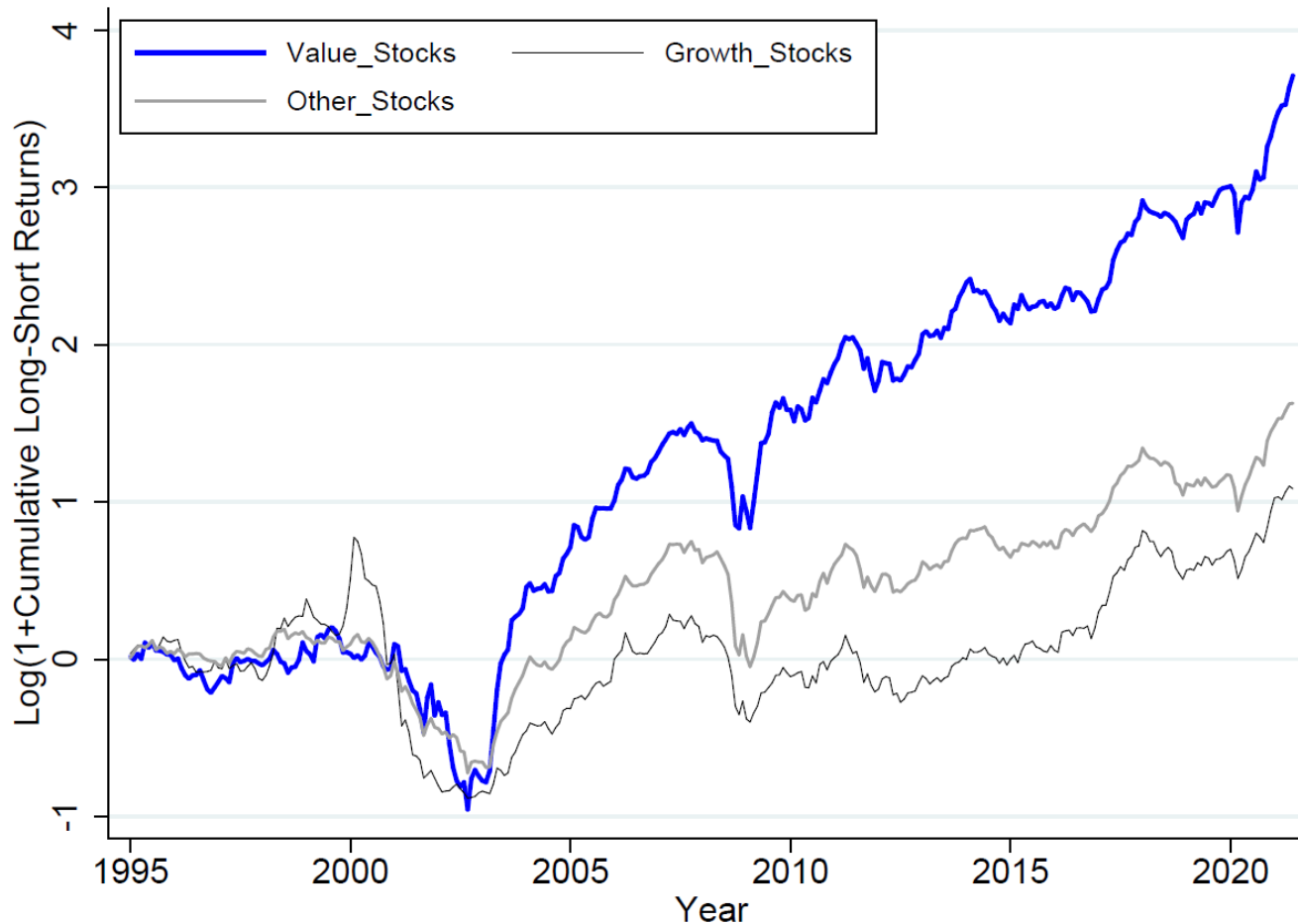
- **Shareholder's equity: Book value vs. market value**

- **Market-to-book ratio (= Price-to-book (P/B) ratio)**

$$\text{Market-to-book (M/B) ratio} = \frac{\text{Market value of equity}}{\text{Book value of equity}}$$

- Financial ratio to evaluate a firm: Successful firms have a M/B ratio of greater than one.
- Value Stocks: Low M/B ratios
- Growth stocks: High M/B ratios

- Invest in 10% stocks with lowest M/B ratio (Value Stocks)
- Short 10% stocks with highest M/B ratio (Growth Stocks)



Return p.a.

Value: 17.2%

Growth: 6.4%

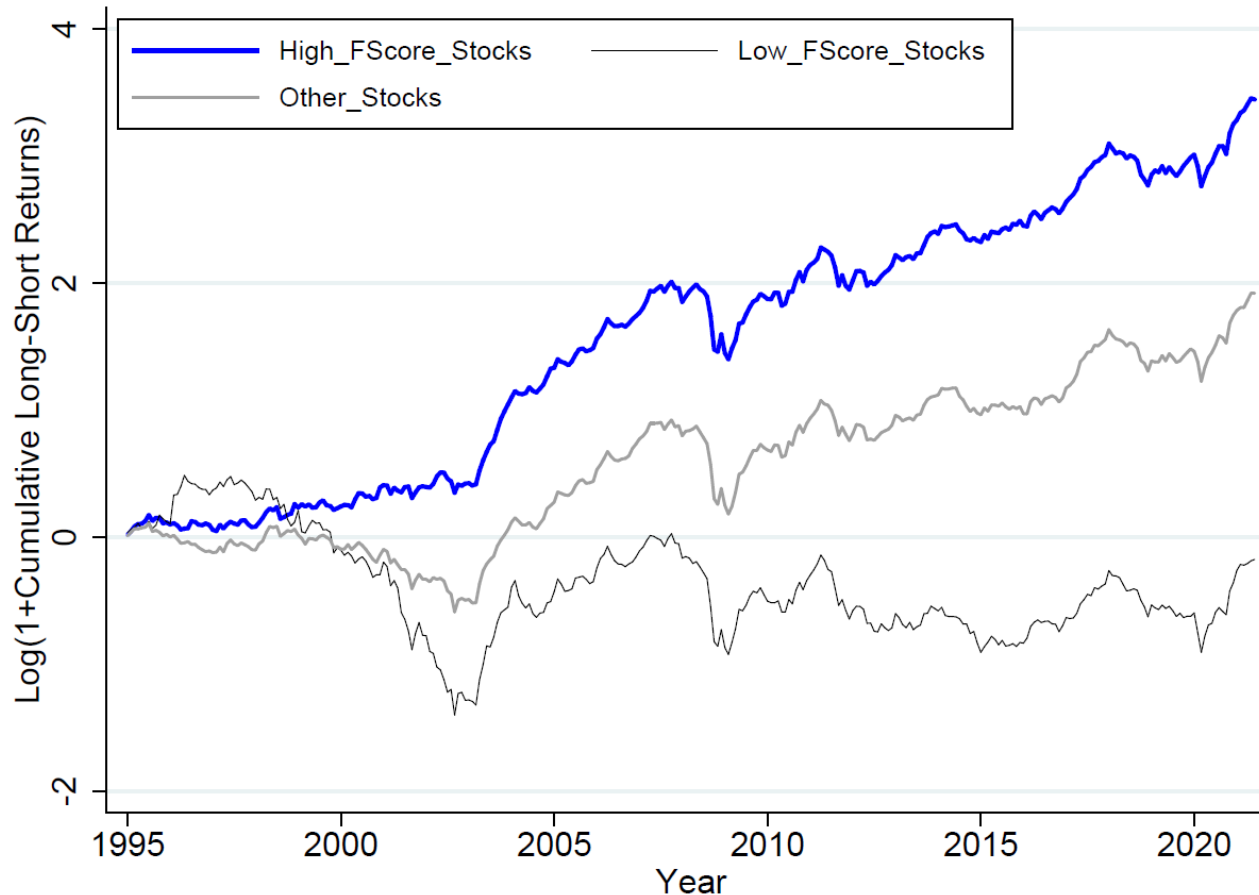
- Financial strength signal based on 9 binary indicators on profitability, leverage/liquidity, and operating efficiency
- Ranges from 0 to 9, with 9 indicating best (most healthy) financial statements

PROFITABILITY	LEVERAGE, LIQUIDITY, AND SOURCE OF FUNDS	OPERATING EFFICIENCY
Positive net income	Lower ratio of long term debt in the current period, compared to the previous year	A higher gross margin compared to the previous year
Positive return on assets in the current year	Higher current ratio this year compared to the previous year	A higher asset turnover ratio compared to the previous year
Positive operating cash flow in the current year	No new shares were issued in the last year	
Cash flow from operations being greater than net income		

Source: <https://tokenist.com/investing/piotroski-f-score/>

¹Piotroski, J. D., 2000, "Value investing: The use of historical financial statement information to separate winners from losers," Journal of Accounting Research, 38, 1-41.

- Invest in stocks with F-Score ≥ 8
- Short stocks with F-Score ≤ 3



Return p.a.

High F-Score: 14.4%

Low F-Score: 2.0%

- **The strategies work (much) better for smaller stocks**
- **You cannot short-sell all stocks & have to pay short-selling fees**
- **Frequent portfolio rebalancing necessary (→ higher transaction costs)**
- **Over time strategies became less successful (at least in other markets)**
- **It could be risk (but why should stocks with better balance sheets be riskier?)**
- **Strategies based on single variables are inefficient (you can do better)**